Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_baseline_heuristic_only.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_013822_g
- started_at_local: 2026-05-20 01:38:22
- finished_at_local: 2026-05-20 02:10:40
- duration_hhmm: 00:32
- duration_seconds: 1938.115
- seed_offset: 6000
- replicate_label: g
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.880405	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.054968	0.121317	0.752867	0.4475	False	0.002607	0.005797
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235667	0.054968	0.121542	0.903553	0.445	False	0.00807	0.562244
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235667	0.054968	0.121542	0.844238	0.4475	False	0.002904	0.179435

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.054968	0.121317	0.0	0.44	False	0.002151	0.033207
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.238386	0.064041	0.128273	0.722793	0.4425	False	0.003018	0.015857

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.880405, complexity 0.56, and adaptation efficiency -0.028288.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.752867, complexity 0.4475, and adaptation efficiency 0.007298.
- Last-epoch runtime 67.9039 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.002607, positive scaffolds 0, Pareto runtime inflation 0.005797.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.005797, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.903553, complexity 0.445, and adaptation efficiency -0.005611.
- Last-epoch runtime 49.282625 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.00807, positive scaffolds 1, Pareto runtime inflation 0.562244.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.404735, "gap_delta": 0.000224, "runtime_inflation": 0.562244, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.844238, complexity 0.4475, and adaptation efficiency 0.009629.
- Last-epoch runtime 202.169275 ms and distance evaluations 6995.75.
- Validation: surviving False, transplant delta 0.002904, positive scaffolds 1, Pareto runtime inflation 0.179435.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.156445, "gap_delta": 0.000224, "runtime_inflation": 0.179435, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.44, and adaptation efficiency 0.0.
- Last-epoch runtime 72.158575 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.002151, positive scaffolds 0, Pareto runtime inflation 0.033207.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.033207, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.238386, family holdout 0.064041, combined 0.128273.
- Accepted novelty 0.722793, complexity 0.4425, and adaptation efficiency 0.006842.
- Last-epoch runtime 134.850475 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.003018, positive scaffolds 0, Pareto runtime inflation 0.015857.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 0.003718, "gap_delta": 0.006956, "runtime_inflation": 0.015857, "same_gap_faster": false}

Judge Appendix

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Summary of TSP Modular-Operator Discovery Results

Key Criteria Recap

- **Main transfer evidence:** held-out TSPLIB gap and family holdout gap.
- **Modular operator interest:** requires validation surviving transplant and ablation tests.
- **Claims:** focus on reusable operator structure over full-solver performance.

- **Algorithm discovery:** requires operator survival in scaffold and family tests.

Overview of Conditions

Condition Name	Execution Mode	Operator Type	Final Family Gap	Final TSPLIB Gap	Transfer Gap	Surviving Candidate	Transplant Mean Gap Delta	Ablation Result	Comments
phase7_baseline_heuristic_only	baseline_only	n/a	0.054968	0.235058	0.121317	No	0.0	N/A	Baseline reference
phase7_full_solver_evolution	full_solver	n/a	0.064916	0.213387	0.159398	No	0.0	N/A	Full solver evolution, no operator
phase7_modular_operator_evolution	modular_operator	perturbation	0.054968	0.235058	0.121317	No	0.002607	No effect	No improvement; ablation disables operator effect
phase7_modular_operator_random_replay	modular_operator	candidate_pruner	0.054968	0.235667	0.121542	No	0.00807	No effect	Runtime inflation high; no positive family gain
phase7_modular_operator_diversity_residual_replay	modular_operator	candidate_pruner	0.054968	0.235667	0.121542	No	0.002904	No effect	Similar to above; no validation gain
phase7_modular_operator_compression_pressure	modular_operator	perturbation	0.054968	0.235058	0.121317	No	0.002151	No effect	Operator disables w/o loss; no transplant success
phase7_modular_operator_pareto_selection	modular_operator	candidate_ranker	0.064041	0.238386	0.128273	No	0.003018	No effect	Underperforms on grid_like_tsp; no transplant success

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Detailed Interpretation

1. Held-Out and Transfer Performance

- **Held-out TSPLIB gaps** remain large (~0.21-0.24) across conditions.
- **Family gaps** mostly unchanged (~0.05-0.06), consistent with the baseline.
- **Transfer gaps** show no meaningful improvement compared to baseline; some operators have marginally negative or zero impact.
- None of the operators demonstrate consistent or statistically significant improvements on held-out TSPLIB or family holdouts.

2. Transplantation Checks

- Mean transplant gap deltas are small and often positive (indicating slight worsening).
- No operator shows a positive transplant scaffold count above zero except candidate pruners in random and residual replay modes (positive scaffold count=1), but gap deltas remain negligible or slightly negative.
- Perturbation operators fail to transplant cleanly into other scaffolds (notably clustered_local_search).
- Candidate ranker does not transplant cleanly anywhere.

3. Ablation Results

- In all modular operators tested, disabling the operator has **zero or negative gap delta**:
- Disabling does not degrade performance, indicating no clear positive operator effect.
- Variants (simplified, alternate, shuffled) do not improve or meaningfully change gaps.
- This undermines claims about real operator effects or reusable structures.

4. Pareto Tradeoffs

- Pareto gap deltas for candidate ranker are positive (worse gaps), and runtime is slightly inflated.
- Runtime inflation for candidate pruners is substantial (up to 56%), with no gap improvement.
- Perturbation operators show negligible runtime inflation (~0-3%) but no gap gains.
- No operator achieves strictly better gap for equal or less runtime.

5. Operator Survival and Reusability

- No condition produces a **surviving candidate operator** per conditions.
- No operator survives all validation phases including ablation, transplant, and family holdout.
- Operators fail transplant tests into multiple scaffolds, limiting reusability.
- Lack of positive family gap improvement undermines reuse claims.

6. Special Notes on Operator Core Ideas

- Perturbations aim at deterministic stagnation escape via double-bridge style moves but fail to improve gaps or survive ablations.
- Candidate pruners adapt neighborhood sizes using instance features but incur runtime inflation without quality gain.
- Candidate ranker prioritizes bottleneck edges but shows performance regressions and fails transplant.

Conclusion

- **No modular operator candidate currently demonstrates convincing transfer or transplant validation.**
- Ablation analysis universally shows disabling operators does not harm performance, thus no confirmed positive modular effect.
- Held-out TSPLIB and family gaps remain at baseline levels without meaningful improvements.
- Runtime tradeoffs are either neutral or negative (inflation without quality gain).
- Operators do not reliably transplant to other scaffolds, limiting claims on reusable operator structures.

- **No algorithm discovery or reusable operator claims are warranted given the lack of survival through ablation, transplant, and family validation.**

Recommendation

- Continue exploration with stronger family holdout and transplant validations.
- Focus on operators showing positive ablation effects and consistent family improvements before claiming reusability or discovery.
- Avoid claiming modular operator effectiveness or solver improvement absent clear validation.